BE-203 · BTech First Year · UIT, Barkatullah University, Bhopal

Fundamentals of Computer Programming — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

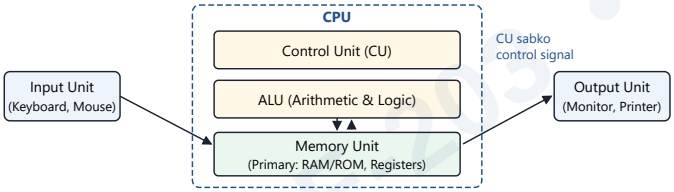
UNIT I Computer Fundamentals, Operating System & MS Office

1. Computer – Definition

DEFINITION **Computer** ek electronic device hai jo user se **data (input)** leta hai, stored **instructions (program)** ke according use **process** karta hai, aur **information (output)** deta hai; result ko future use ke liye **store** bhi kar sakta hai.

Characteristics: Speed (nano/pico seconds), Accuracy (GIGO – Garbage In Garbage Out), Diligence (thakta nahi), Versatility, Storage capacity, Automation. **Limitation:** No IQ, no feelings – khud decision nahi leta.

2. Block Schematic of Digital Computer & Working



Input → Memory ⇄ ALU (CU sab control karta hai) → Output

- **Input Unit:** data ko binary mein convert karke memory tak bhejta hai.
- **Memory Unit:** data, program, intermediate results store (primary RAM/ROM/cache; secondary HDD/SSD).
- **ALU:** arithmetic (+ − × ÷) aur logical (AND, OR, NOT, compare) operations.
- **Control Unit:** instructions ka **fetch** → **decode** → **execute**; sab units ko control signals.
- **Output Unit:** result ko human-readable form mein deta hai.

Yaad Rakho: CPU = CU + ALU + Registers. Working cycle = **IPOS**: Input → Process → Output → Storage; **Von Neumann architecture** (stored program concept).

3. Classification of Computers

- **By working/data: Analog** (continuous data – thermometer, speedometer), **Digital** (discrete 0/1 – PC), **Hybrid** (dono – ICU machine, petrol pump).
- **By size & power: Micro** (PC/laptop), **Mini** (small org, multi-user), **Mainframe** (banks, railways – IBM z), **Super** (weather/research – PARAM, Frontier; speed in FLOPS).
- **By purpose: General purpose** (PC) aur **Special purpose** (ATM, washing machine controller, traffic light).

4. Generations of Computer

Gen	Period	Technology	Language	Example
1st	1940–56	Vacuum tubes; magnetic drum	Machine language	ENIAC, EDVAC, UNIVAC
2nd	1956–63	Transistors; magnetic core	Assembly, early FORTRAN/COBOL	IBM 1401, IBM 7090
3rd	1964–71	Integrated Circuits (IC)	High-level (BASIC, Pascal); OS aaya	IBM 360, PDP-8
4th	1971–present	Microprocessor (VLSI)	C, C++, GUI OS	IBM PC, Apple Macintosh
5th	Present–future	AI, ULSI, parallel processing	Python, Prolog, NLP	Robots, AI assistants

5. I/O Devices & Storage Devices

Input: Keyboard, Mouse, Scanner, Joystick, Light pen, Mic, Webcam, Touch screen, Barcode reader, **OCR** (Optical Character Reader), **MICR** (bank cheques), **OMR** (exam OMR sheet), Biometric. **Output:** Monitor (CRT/LCD/LED), Printer (Impact – dot matrix; Non-impact – inkjet, laser), Plotter, Speaker, Projector. Soft copy = screen, Hard copy = paper.

Primary Memory	Secondary Memory
CPU directly access karta hai; fast, costly, small. **RAM** (volatile): SRAM (cache), DRAM (main). **ROM** (non-volatile): PROM, EPROM (UV erase), EEPROM (electrical erase) – BIOS	CPU directly access nahi; slow, cheap, large, non-volatile: HDD (magnetic), SSD/Pen drive (flash), CD/DVD/Blu-ray (optical), Magnetic tape (sequential). 1 Byte = 8 bits; 1 KB = 1024 B → MB, GB, TB

Memory hierarchy (fast → slow): Registers → Cache → RAM → SSD/HDD → Optical/Tape (speed ↓, size ↑, cost/bit ↓).

6. System Software vs Application Software

System Software	Application Software
Hardware manage karta hai, platform deta hai; iske bina computer nahi chalega. Ex: OS, Compiler, Interpreter, Assembler, Loader, Linker, Device drivers, Utilities (antivirus)	User ka specific task karta hai; system software par depend. Ex: MS Word, Excel, Tally, browser, games, Photoshop

7. Computer Applications

e-Business: online buying/selling (B2B, B2C), e-banking, UPI, online marketing, ERP (Amazon, Flipkart).

Bio-Informatics: DNA/protein sequence data ka storage & analysis; genome sequencing, drug design (BLAST tool).

Health Care: hospital management, e-health records, CT/MRI imaging, telemedicine, robotic surgery.

Remote Sensing & GIS: satellite se bina contact earth ka data; GIS spatial data analyse karta hai – maps, urban planning, disaster & crop monitoring.

Meteorology & Climatology: supercomputers par numerical weather models – forecasting, cyclone tracking, climate change study (IMD).

Computer Gaming: video games, GPU graphics, VR/AR, simulators (flight training), e-sports.

Multimedia & Animation: text + audio + video + image + animation combine; 2D/3D films, e-learning, ads (Maya, Blender).

8. Operating System

> **DEFINITION** **Operating System (OS)** ek system software hai jo **user aur hardware ke beech interface** ka kaam karta hai aur computer ke saare resources (CPU, memory, I/O, files) ko manage karta hai. Ex: Windows, UNIX, Linux, macOS, Android.

Functions of OS: Process management, Memory management, File management, Device (I/O) management, Security & protection, User interface (CLI/GUI), Job accounting, Error detection, Booting.

Types of OS:

Batch OS: similar jobs ka batch, bina user interaction (payroll).

Multiprogramming: memory mein many programs; CPU idle nahi.

Time-sharing / Multitasking: har process ko time-slice (quantum); multi-user (UNIX).

Real-Time OS: fixed deadline mein response; Hard/Soft (missile, ABS brakes, VxWorks).

Distributed OS: many networked computers ek system jaise (Amoeba).

Network OS: server se file/printer sharing (Windows Server, NetWare).

Mobile / Single-user OS: Android, iOS / MS-DOS.

File, Process & Memory Management

- **File Management:** files/directories create, delete, read, write, rename; hierarchical directory structure; access permissions; disk space allocation (contiguous, linked, indexed).
- **Process Management:** Process = program in execution. States: **New** → **Ready** → **Running** → **Waiting** → **Terminated**. OS CPU scheduling karta hai (FCFS, SJF, Round Robin, Priority), PCB (Process Control Block) maintain karta hai, deadlock handle karta hai.
- **Memory Management:** process ko memory allocate/deallocate karna; track karna kaun sa part free hai; techniques – partitioning, **paging**, **segmentation**, **virtual memory** (RAM kam ho to disk ka part RAM ki tarah use).

9. Introduction to UNIX and Windows

UNIX	Windows
1969, AT&T Bell Labs (Ken Thompson, Dennis Ritchie); C mein likha. Multi-user, multitasking, portable, secure; mainly CLI. Structure: Hardware → **Kernel** → **Shell** → Utilities. Case-sensitive; variants Linux, BSD	Microsoft (1985 se), GUI based (icons, windows, menus); multitasking. Desktop, Taskbar, Start menu, File Explorer, Control Panel; NTFS file system; proprietary (Windows 10/11)

Simple UNIX Commands & Directory Commands

Command	Kaam	Command	Kaam
ls, ls -l	files list (long format)	pwd	present working directory
cat f1 / cat > f1	file display / create	mkdir d1, rmdir d1	dir banao / empty dir delete
cp f1 f2, mv f1 f2	copy, move/rename	cd d1, cd .., cd ~	dir change / parent / home
rm f1 (rm -r dir)	delete	chmod 755 f1	permissions (rwx)
date, cal, who, man ls	date, calendar, users, help	wc, grep "x" f1, head/tail	count, search, first/last lines

10. Introduction to MS Word, PowerPoint, Excel

Software	Use	Important Features
MS Word (.docx)	Word processor – letters, reports, resume	Formatting (font, bold, alignment), tables, header/footer, page numbers, spell-check, **Mail Merge**, Find & Replace, Track changes
MS PowerPoint (.pptx)	Presentation – slides	Slide layouts, themes, **animations** (object par), **transitions** (slide change), Slide Master, Slide Show (F5)
MS Excel (.xlsx)	Spreadsheet – calculations, data analysis	Workbook → Worksheets → Cells (A1 address); formulas start with **=**: =SUM(A1:A5), =AVERAGE(), =MAX(), =IF(); charts, sorting, filtering

UNIT II Algorithm, Programming Concepts & Introduction to C

1. Algorithm

> **DEFINITION** **Algorithm** = kisi problem ko solve karne ke liye **finite, well-defined, step-by-step instructions** ka sequence. Properties: **Input** (0 ya more), **Output** (≥1), **Definiteness** (har step clear), **Finiteness** (finite steps mein khatam), **Effectiveness** (simple, doable).

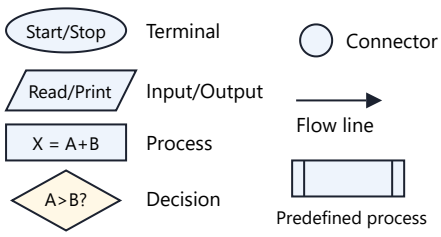
Example – Largest of two numbers: Step 1: Start. Step 2: Read A, B. Step 3: If A > B print A else print B. Step 4: Stop.

Complexity of Algorithm

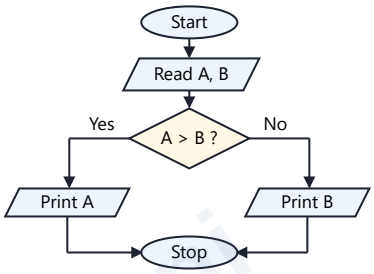
- **Time complexity:** input size n ke function mein algorithm kitna time (steps) leta hai. **Space complexity:** kitni memory leta hai.
- Asymptotic notations: **Big-O** (upper bound / worst case), **Ω Omega** (lower bound / best case), **Θ Theta** (tight bound / average).
- Order: O(1) < O(log n) < O(n) < O(n log n) < $O(n^2)$ < $O(2^n)$. Ex: Linear search O(n), Binary search O(log n), Bubble sort $O(n^2)$.

Flowchart

Flowchart = algorithm ka **graphical/pictorial representation** standard symbols ke saath.



Standard flowchart symbols



Flowchart: largest of two numbers

2. Introduction to Programming & Categories of Languages

Program = instructions ka set jo computer ko task karne ke liye diya jata hai. **Programming** = program likhne ka process. **Translators:** Assembler (assembly → machine), Compiler (poora HLL program ek baar mein translate – C), Interpreter (line by line – Python, BASIC).

Category	Description	Pros / Cons
Machine (1GL) – low level	Binary 0/1 code, CPU directly samajhta hai	Fastest / machine dependent, error-prone
Assembly (2GL) – low level	Mnemonics (ADD, MOV); assembler chahiye	Easier than binary / machine dependent
High-Level (3GL)	English-like (C, C++, Java, FORTRAN)	Portable, easy / translation needed
4GL / 5GL	Non-procedural (SQL, MATLAB) / AI (Prolog, LISP)	"What" batao, "how" nahi

3. Program Design & Programming Paradigms

Program Design (Program Development Life Cycle): Problem definition → Analysis → Design (algorithm/flowchart/pseudocode) → Coding → Testing & Debugging → Documentation → Maintenance. Approaches: **Top-down** (bade problem ko chhote modules mein todna – C), **Bottom-up** (chhote objects se system banana – C++/Java), **Modular** programming.

Programming Paradigms (programming ki style/approach): **Procedural/Structured** – program = functions ka sequence, top-down (C, Pascal); **Object-Oriented** – program = objects (data + functions), bottom-up (C++, Java); **Functional** – mathematical functions, no side effects (Haskell, LISP); **Logic/Declarative** – facts + rules (Prolog, SQL); **Event-driven** – click/key events par code chale (Visual Basic, JavaScript).

4. OOP – Characteristics/Concepts & POP vs OOP

Class: blueprint/template – data members + member functions (ex: class Car).

Object: class ka instance (real entity) – ex: Car c1;

Encapsulation: data + functions ko ek unit (class) mein wrap karna.

Abstraction: sirf essential features dikhana, details chhupana (ATM use karna).

Data Hiding: private data bahar se access nahi.

Inheritance: ek class doosri class ki properties le (reusability) – parent → child.

Polymorphism: "one name, many forms" – function/operator overloading, overriding.

Dynamic Binding: kaunsa function call hoga ye run-time par decide.

Message Passing: objects function call se communicate karte hain.

POP (Procedure Oriented)	OOP (Object Oriented)
Focus on functions; top-down approach	Focus on data (objects); bottom-up approach
Global data freely move karta hai – less secure	Data hiding via access specifiers (private, public, protected)
No inheritance/overloading; bade program maintain karna mushkil	Inheritance, polymorphism; reusable, real-world modelling
Ex: C, Pascal, FORTRAN	Ex: C++, Java, C#, Python

5. Introduction to C – Program Structure

C language **Dennis Ritchie** ne **1972** mein AT&T Bell Labs par develop ki (UNIX likhne ke liye). C = middle-level, structured, portable, fast, case-sensitive language.

```
/* Documentation section */
#include <stdio.h>   // Link section
#define PI 3.14       // Definition section
int count = 0;        // Global declaration
int main() {          // main() function
    int r = 2; float area;   // declaration
    area = PI * r * r;       // executable part
    printf("Area = %f\n", area);  return 0;
}
/* User-defined functions (subprogram) */
```

Sections: Documentation (comments) → Link (#include header files) → Definition (#define constants) → Global declaration → **main()** (execution yahi se start) → Sub-program (user functions).

Compile process: Source (.c) → Preprocessor → Compiler → Object (.obj/.o) → Linker → Executable (.exe).

Tokens: Keywords (32 in C89: int, if, while, return...), Identifiers, Constants, Strings, Operators, Special symbols.

Data Types

Type	Size (typical 32/64-bit GCC)	Format specifier	Range / Example
char	1 byte	%c	−128 to 127; 'A'
int	4 bytes (Turbo C: 2 bytes)	%d	$\pm 2.1 \times 10^9$ approx (2-byte: −32768 to 32767)
float	4 bytes	%f	~6 decimal digits precision; 3.14
double	8 bytes	%lf	~15 digits precision

void = no value. Modifiers: short, long, signed, unsigned. **Derived:** array, pointer, function. **User-defined:** struct, union, enum, typedef. **Assignment statement:** `x = 10;` Compound: `x += 5;` (x = x + 5), `-=, *=, /=, %=`. Multiple: `a = b = c = 0;` (right to left).

6. Operators – Unary, Binary, Ternary

Type	Operators	Example
Unary (1 operand)	++, --, unary −, !, ~, sizeof, & (address), * (value at)	++a (pre: pehle badhao phir use), a++ (post: pehle use phir badhao)
Binary – Arithmetic	+ − * / %	7/2 = 3 (int division), 7%2 = 1
Binary – Relational / Logical	< > <= >= == != / && ||	result 1 (true) ya 0 (false); (a>0 && b>0)
Binary – Bitwise / Assignment	& | ^ << >> / = += −= *= /= %=	5&3 = 1, 5|3 = 7, 5^3 = 6, 5<<1 = 10; a += 2
Ternary (3 operands)	condition ? expr1 : expr2	max = (a > b) ? a : b;

COMMON MISTAKE = (assignment) aur == (comparison) mein confusion. `if(x = 5)` hamesha true hoga! a=5: `b = a++;` → b=5, a=6; `b = ++a;` → b=6, a=6.

Precedence (high → low): () [] → Unary → * / % → + − → << >> → < <= > >= → == != → & → ^ → | → && → || → ?: → = += ...

7. Input-Output Statements & Simple C Programs

Formatted: `scanf("%d", &x);` (address operator & zaroori), `printf("%d", x);`. **Unformatted:** getchar(), putchar(), gets()/fgets(), puts(), getch().

```c
/* Sum and average of 3 numbers */
#include <stdio.h>
int main() {
    int a, b, c;  float avg;
    scanf("%d %d %d", &a, &b, &c);
    avg = (a + b + c) / 3.0;
    printf("Sum=%d Avg=%.2f\n", a+b+c, avg);
    return 0;
}
```

```c
/* Swap without third variable */
int a = 10, b = 20;
a = a + b;   /* a = 30 */
b = a - b;   /* b = 10 */
a = a - b;   /* a = 20 */
/* Area of circle: area = 3.14*r*r; */
/* C to F: f = c * 9.0 / 5 + 32;    */
```

8. Decision Making & Branching

if Statements

Types: simple **if**, **if-else**, **nested if**, **else-if ladder**.

```c
/* if-else : even or odd */
if (n % 2 == 0) printf("Even");
else            printf("Odd");
/* else-if ladder : grade */
if (m >= 75)      printf("A");
else if (m >= 60) printf("B");
else if (m >= 33) printf("C");
else              printf("Fail");
```

```c
/* nested if : largest of three */
if (a > b) {
    if (a > c) printf("%d", a);
    else       printf("%d", c);
} else {
    if (b > c) printf("%d", b);
    else       printf("%d", c);
}
```

switch Statement

```c
switch (op) {
  case '+': r = a + b; break;
  case '-': r = a - b; break;
  case '*': r = a * b; break;
  default : printf("Invalid");
}
```

Expression ka value **int ya char** hona chahiye (float/string nahi). **break** na ho to **fall-through** – agle cases bhi execute. **default** optional hai. Duplicate case labels allowed nahi.

Loops – for, while, do-while

for	while	do-while
for(init; cond; update)	while(cond) { ... }	do { ... } while(cond);
Entry-controlled; jab iterations pata ho	Entry-controlled; condition pehle check	**Exit-controlled**; body **kam se kam ek baar** chalegi

```c
/* Factorial using for */
long f = 1;
for (i = 1; i <= n; i++)  f = f * i;
/* Reverse digits using while */
while (n != 0) {
    rev = rev * 10 + n % 10;  n = n / 10;
}
```

```c
/* Sum of digits using do-while */
do { sum += n % 10;  n /= 10; } while (n > 0);
/* break & continue */
for (i = 1; i <= 10; i++) {
    if (i == 3) continue;  /* skip 3    */
    if (i == 6) break;     /* stop at 6 */
    printf("%d ", i);      /* 1 2 4 5   */
}
```

break = loop/switch se turant bahar; **continue** = current iteration skip, next par jao; **goto** = label par jump (avoid). Practice: prime, Fibonacci, palindrome, Armstrong number.

UNIT III Arrays, Functions, Structures, Recursion, Pointers, Files

1. Arrays

> **DEFINITION** **Array** = same data type ke elements ka collection jo **contiguous memory locations** mein store hota hai aur ek common name + **index** (0 se start) se access hota hai.

Declaration: `int a[5] = {10, 20, 30, 40, 50};` → a[0]=10 ... a[4]=50. 2-D: `int m[3][3];` (row-major order mein store). String = char array ending with `'\0'`: `char s[] = "UIT";` (size 4). String functions (<string.h>): strlen, strcpy, strcat, strcmp, strrev (non-standard, Turbo C).

```c
/* Largest element of array */
int a[5] = {4, 9, 2, 7, 5}, i, max = a[0];
for (i = 1; i < 5; i++)
    if (a[i] > max) max = a[i];
printf("Max = %d", max);   /* 9 */
```

```c
/* Matrix addition (2-D array) */
for (i = 0; i < 2; i++)
  for (j = 0; j < 2; j++)
    c[i][j] = a[i][j] + b[i][j];
/* Address of a[i] = Base + i*sizeof(type) */
```

2. Functions

Function = self-contained block of code jo specific task karta hai. Benefits: modularity, reusability, easy debugging. Types: **Library** (printf, sqrt, strlen) aur **User-defined**. Parts: **declaration (prototype)**, **definition**, **call**.

```c
#include <stdio.h>
void swap(int *x, int *y);  /* prototype */
int main() {
    int a = 5, b = 8;
    swap(&a, &b);           /* call by reference */
    printf("%d %d", a, b);  /* 8 5 */
    return 0;
}
void swap(int *x, int *y) {  /* definition */
    int t = *x; *x = *y; *y = t;
}
```

Call by Value	Call by Reference
Value ki copy pass hoti hai	Address (pointer) pass hota hai
Original variable change nahi hota	Original variable change hota hai
swap(a, b)	swap(&a, &b)

Categories: no arg-no return, arg-no return, no arg-return, arg-return.

3. Structures (and Union)

Structure = different data types ke related items ka user-defined group. Members ko **dot (.)** operator se access karte hain; pointer se **arrow (->)**.

```c
struct student { int roll; char name[20];
                 float marks; };
struct student s1 = {1, "Ravi", 82.5};
printf("%d %s", s1.roll, s1.name);
struct student *p = &s1;  printf("%d", p->roll);
```

Structure	Union
Har member ki alag memory	Sab members ek hi memory share
Size ≥ sum of members	Size = largest member
Sab members ek saath use (struct)	Ek time par ek member valid (union)

4. Recursion

Jab function **khud ko call** kare = recursion. Har recursive function mein **base condition** (termination) zaroori, warna infinite calls → stack overflow. Internally **stack** use hota hai.

```c
/* Factorial: n! = n * (n-1)! */
long fact(int n) {
    if (n == 0 || n == 1) return 1;  /* base */
    return n * fact(n - 1);
}
/* fact(4) = 4*3*2*1 = 24 */
```

```c
/* Fibonacci: 0 1 1 2 3 5 8 ... */
int fib(int n) {
    if (n <= 1) return n;         /* base */
    return fib(n - 1) + fib(n - 2);
}
```

Recursion vs Iteration: recursion – code chhota, elegant, par zyada memory (stack) aur slow; iteration – loop based, fast, kam memory.

5. Pointers

> **DEFINITION** **Pointer** ek variable hai jo doosre variable ka **memory address** store karta hai. & = address-of operator, * = dereference (value at address).

```c
int x = 10;
int *p = &x;   /* p stores address of x */
printf("%d %d", x, *p);  /* 10 10 */
printf("%p", (void*)p);  /* address of x */
*p = 20;            /* x becomes 20 */
int **q = &p;       /* pointer to pointer */
```

- **Pointer arithmetic:** p++ next element par jata hai (sizeof(type) bytes aage).
- **Array & pointer:** a == &a[0]; a[i] ≡ *(a + i).
- **NULL pointer:** kuch point nahi karta (int *p = NULL;).
- **void pointer:** generic, kisi bhi type ka address.
- **Dangling pointer:** free ki hui memory ko point kare.
- Dynamic memory: malloc(), calloc(), realloc(), free() (<stdlib.h>).

6. Files and File Handling

File = secondary storage par permanent data. Program khatam hone ke baad bhi data rehta hai. `FILE *fp;` pointer use hota hai (<stdio.h>). Types: **Text file** aur **Binary file**.

Mode	Meaning	Function	Kaam
"r"	read (file honi chahiye, warna NULL)	fopen(), fclose()	open / close
"w"	write (nayi banegi / purana data erase)	fprintf(), fscanf()	formatted write/read
"a"	append (end mein add)	fputc(), fgetc()	character write/read
"r+", "w+", "a+"	read + write	fputs(), fgets()	string write/read
"rb", "wb"	binary mode	fread(), fwrite(), fseek(), ftell(), rewind(), feof()	block I/O, positioning

```
/* Write to a file */
FILE *fp = fopen("data.txt", "w");
if (fp == NULL) { printf("Error"); return 1; }
fprintf(fp, "Roll=%d\n", 101);
fclose(fp);
```

```
/* Read file character by character */
FILE *fp = fopen("data.txt", "r");  int ch;
while ((ch = fgetc(fp)) != EOF) putchar(ch);
fclose(fp);
```

7. Storage Classes & Scope of Variables

Storage Class	Storage	Default value	Scope	Lifetime
auto (default local)	Memory (stack)	Garbage	Block (local)	Block khatam tak
register	CPU register (agar free ho)	Garbage	Block (local)	Block khatam tak; & address nahi le sakte
static	Memory (data segment)	Zero	Block (local) / file (global static)	Poore program tak – value calls ke beech retain
extern	Memory	Zero	Global (multiple files)	Poore program tak

```
void count() {
    static int c = 0;  /* sirf ek baar init */
    c++;  printf("%d ", c);
}
/* count() 3 baar call: output 1 2 3 */
/* agar 'auto int c' hota: 1 1 1 */
```

Scope of variables: Local – function/block ke andar declare, wahin visible. **Global** – sab functions ke bahar declare, poore program mein visible. **Block scope** { }, **function scope** (labels), **file scope**. Same naam ho to local, global ko **hide** kar deta hai.

8. Basic Concepts of OOPs (Revision)

OOP ke basic concepts: **Class, Object, Encapsulation, Abstraction, Data hiding, Inheritance, Polymorphism, Dynamic binding, Message passing** (details Unit II mein). C procedural hai; OOP **C++** mein implement hota hai. C ka **struct** sirf data rakhta hai, C++ **class** data + functions + access control rakhti hai – ex: `class Rect { private: int l, b; public: int area() { return l*b; } };` aur `Rect r1;` (r1 = object).

UNIT IV Computer Networking & Computer Security

1. Computer Networking – Introduction

> **DEFINITION** **Computer network** = do ya zyada computers/devices ka interconnection (wired ya wireless) jisse wo **data, resources (printer, files) aur information share** kar sakein.

Types by area: PAN (~10 m, Bluetooth), **LAN** (building/campus, Ethernet), **MAN** (city, cable TV), **WAN** (country/world, Internet). **Topologies:** Bus, Star (central hub/switch), Ring, Mesh (n(n−1)/2 links), Tree, Hybrid. **Media:** Twisted pair, Coaxial, Optical fibre (guided); Radio, Microwave, Infrared, Satellite (unguided).

2. ISO-OSI Reference Model (7 Layers)

#	Layer	Main Function	Data unit / Device / Protocol
7	Application	User ko network services (email, browsing)	HTTP, FTP, SMTP, DNS
6	Presentation	Translation, encryption, compression	JPEG, SSL/TLS, ASCII
5	Session	Session establish, maintain, terminate; synchronization	RPC, NetBIOS
4	Transport	End-to-end delivery, segmentation, flow & error control	Segment; TCP, UDP
3	Network	Logical addressing (IP), routing	Packet; Router; IP
2	Data Link	Framing, physical (MAC) addressing, error detection	Frame; Switch, Bridge
1	Physical	Bits ka transmission, cables, signals	Bits; Hub, Repeater

Yaad Rakho (1→7): "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way". OSI ISO ne 1984 mein publish kiya.

3. Internetworking Concepts & Devices

Internetworking = alag-alag networks (LAN, WAN, different technologies) ko devices (router, gateway) se jodkar ek bada network banana – jaise **Internet** = "network of networks".

Device	OSI Layer	Kaam
Repeater / Hub	Physical	Repeater: weak signal regenerate; Hub: data sab ports par broadcast (dumb)
Bridge / Switch	Data Link	MAC address se filter; switch sirf destination port par bhejta hai (smart hub)
Router	Network	Different networks jodta hai, IP address se best path (routing)
Gateway	All layers	Different protocols wale networks jodta hai (protocol converter)
Modem / NIC	Physical / DL	Modem: digital ⇄ analog; NIC: network card with unique 48-bit MAC

4. TCP/IP Model

TCP/IP Layer	OSI equivalent	Protocols
Application	Application + Presentation + Session	HTTP, FTP, SMTP, DNS, Telnet
Transport	Transport	TCP, UDP
Internet	Network	IP, ICMP, ARP
Network Access (Link)	Data Link + Physical	Ethernet, Wi-Fi

OSI	TCP/IP
7 layers; theoretical reference model	4 layers; practical, Internet par use
Model pehle, protocols baad mein	Protocols pehle, model baad mein
Strict layering	Loosely layered

TCP = connection-oriented, reliable; **UDP** = connectionless, fast (video, DNS).

5. Internet, WWW & Computer Communication

- **Internet:** worldwide network of networks, TCP/IP use karta hai; origin ARPANET (1969). Services: email, WWW, FTP, chat, video calling, e-commerce. **IP address** (IPv4 = 32-bit, e.g. 192.168.1.1; IPv6 = 128-bit). **DNS** domain name ko IP mein convert karta hai. **ISP** = Internet Service Provider (BSNL, Jio).
- **WWW (World Wide Web):** Tim Berners-Lee (1989, CERN). Internet par hyperlinked web pages ka collection; **HTTP/HTTPS** protocol, **HTML** language, **URL** address (https://www.bubhopal.ac.in), **browser** (Chrome, Firefox), **web server**, search engine (Google). **Internet ≠ WWW** – WWW internet ki ek service hai.
- **Computer Communication:** components – **Sender, Receiver, Message, Medium, Protocol**. Modes: **Simplex** (one-way – keyboard, TV), **Half-duplex** (dono taraf, ek time ek – walkie-talkie), **Full-duplex** (dono taraf ek saath – phone). Bandwidth (bps), Analog vs Digital signal, Serial vs Parallel transmission.

6. Computer Security Basics – Malicious Software

Threat	Meaning	Key feature
Malware	Malicious software ka general naam	Virus, worm, trojan, spyware, ransomware
Virus	Files/programs se attach hokar failta hai	Host program + user action chahiye
Worm	Network se khud self-replicate karta hai	Host nahi chahiye (WannaCry, ILOVEYOU)
Trojan Horse	Useful software jaisa dikhe, andar malicious code	Self-replicate nahi; backdoor kholta hai
Spyware	Chupke se activity/passwords collect kare	Keylogger, adware
Anti-spyware	Spyware detect, block, remove kare	Windows Defender, Malwarebytes

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7. Cyber Attacks

Money Laundering (cyber): illegal (black) money ko online transactions, fake accounts, crypto, shell companies se legal dikhana.

Email Spoofing: email ka "From" address fake karna taaki lage trusted source se aaya hai; phishing ke liye use.

DoS (Denial of Service): server ko fake requests se flood karna taaki genuine users ko service na mile. **DDoS** = many computers (botnet) se.

Cyber Stalking: internet/email/social media se kisi ko baar-baar harass, follow ya threaten karna.

Logic Bomb: program mein chhupa code jo specific condition/date aane par trigger hokar damage kare (ex: employee ke nikaalne par data delete).

Hacking: bina permission computer/network mein unauthorized access. White hat (ethical), Black hat (criminal), Grey hat.

Spamming: bulk unsolicited (bina maange) emails/messages bhejna – ads, scams, malware links.

Phishing: fake website/email se password, OTP, card details churana.

8. Security Measures, Firewall, Ethics, Cyber Laws

- **Security measures:** strong passwords, **2-factor authentication**, antivirus/anti-spyware, **encryption** (plain text → cipher text), regular **backup**, software updates/patches, access control, digital signature, biometric.
- **Firewall:** hardware/software security system jo **incoming & outgoing network traffic** ko predefined rules ke according monitor aur filter karta hai – trusted internal network aur untrusted internet ke beech "wall". Types: packet filtering, stateful inspection, proxy (application gateway).
- **Computer Ethics & Good Practices:** doosre ki files/privacy respect karo, piracy na karo (licensed software), plagiarism na karo, copyright/IPR respect, kisi ko harm na karo, fake news share na karo (Ten Commandments of Computer Ethics).
- **Cyber Laws – IT Act 2000 (India):** 17 Oct 2000 se lagu; **electronic records & digital signatures** ko legal recognition, e-commerce ko support, cyber crimes ki punishment. Amendment 2008. Important sections:

Section	Offence	Section	Offence
43	Unauthorised access/damage – compensation	66C	Identity theft (password misuse)
65	Tampering with source code	66D	Cheating by personation (online fraud)
66	Computer related offences (hacking)	66E / 67	Privacy violation / obscene content

- **Good Security Habits:** unknown links/attachments na kholo, OTP/password share na karo, public Wi-Fi par banking avoid, HTTPS (lock icon) check, logout karo, screen lock, software update, alag-alag passwords, regular backup.

UNIT V Database Management System & Cloud Computing

1. DBMS – Introduction

DEFINITION **Data** = raw facts; **Database** = logically related data ka organized collection; **DBMS** = software jo database ko create, store, retrieve, update, manage karne deta hai (MySQL, Oracle, MS Access).

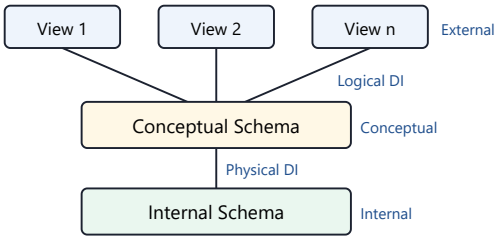
2. File-Oriented Approach vs Database Approach

File-Oriented System	Database Approach (DBMS)
Data **redundancy** (same data many files mein) → **inconsistency**	Redundancy controlled (centralized) → consistency
Har query ke liye naya program; data isolation	Query language (SQL) se easy access, data sharing
Poor security, no concurrency control, no backup/recovery	Security, integrity constraints, concurrency control, backup & recovery
Cheap, simple	Costly, complex

3. Data Models – data, relationships aur constraints describe karne ka conceptual tool

Model	Structure	Note
Hierarchical	Tree (parent-child), 1:N; child ka ek hi parent	IBM IMS
Network	Graph; record ke multiple parents (M:N)	CODASYL, IDMS
Relational	Tables – rows (tuples) & columns (attributes)	E.F. Codd (1970); MySQL, Oracle
E-R Model	Entity (rectangle), Attribute (ellipse), Relationship (diamond)	P. Chen (1976); DB design
Object-Oriented	Data objects + methods	Multimedia, CAD

4. Architecture of Database System (Three-Schema) & Data Independence



ANSI/SPARC three-schema architecture

- **External level (View):** individual user ka view – sirf required data dikhta hai.
- **Conceptual level (Logical):** poore database ka logical structure – entities, relationships, constraints (DBA define karta hai).
- **Internal level (Physical):** data disk par actually kaise store hai – file organisation, indexes.
- **Data Independence** = lower level schema change ho to upper level change na karna pade:
- **Logical data independence:** conceptual schema change (naya column add) → external views change nahi.
- **Physical data independence:** internal schema change (storage, index) → conceptual schema same.

5. Data Dictionary, DBA, Primary Key

- **Data Dictionary:** "data about data" (**metadata**) store karta hai – table/attribute names, data types, constraints, users, access rights (system catalog).
- **DBA (Database Administrator):** person jo poore database par central control rakhta hai. Roles: schema definition, storage & access methods, users ko **authorization**, integrity constraints, **backup & recovery**, performance tuning.
- **Keys: Primary Key** = attribute(s) jo table ki har row ko **uniquely identify** kare; **NULL nahi** ho sakta, **duplicate nahi**, ek table mein ek hi primary key (ex: Roll_No). Others: Super, Candidate, Alternate, Composite, **Foreign key** (doosri table ki primary key ko refer kare).

6. DDL and DML

DDL – Data Definition Language	DML – Data Manipulation Language
Database ka **structure (schema)** define/change karta hai	Database ke andar ke **data** ko access/modify karta hai
CREATE, ALTER, DROP, TRUNCATE, RENAME	SELECT, INSERT, UPDATE, DELETE
Auto-commit (rollback nahi hota)	Rollback possible
Others: **DCL** (GRANT, REVOKE), **TCL** (COMMIT, ROLLBACK, SAVEPOINT)	

```
CREATE TABLE Student (Roll INT PRIMARY KEY, Name VARCHAR(20), Marks INT);   -- DDL
INSERT INTO Student VALUES (101, 'Amit', 78);                              -- DML
UPDATE Student SET Marks = 82 WHERE Roll = 101;                            -- DML
SELECT Name, Marks FROM Student WHERE Marks > 60;                          -- DML
```

7. Cloud Computing

> **DEFINITION** **Cloud Computing** = internet ke through computing resources (servers, storage, databases, software, networking) ko **on-demand**, **pay-as-you-go** basis par provide karna. NIST characteristics: **On-demand self-service, Broad network access, Resource pooling, Rapid elasticity, Measured service**. Ex: AWS, Azure.

Cloud Infrastructure: data centre hardware (servers, storage, network) + **virtualization** (hypervisor – ek physical server par many virtual machines) + management software. Front-end (browser) internet se back-end se judta hai.

Service Models

Model	Provider deta hai	User manage karta hai	Example
IaaS – Infrastructure	VMs, storage, network	OS, runtime, apps, data	AWS EC2, Azure VM
PaaS – Platform	Infra + OS + runtime/dev tools	Apps & data	Google App Engine, Heroku
SaaS – Software	Ready-to-use software	Sirf use karo (control: IaaS > PaaS > SaaS)	Gmail, Google Docs, Zoom

Deployment Models

Model	Description	Example
Public	Third-party, sabke liye open; low cost, less control	AWS, Azure, Google Cloud
Private	Ek organization ke liye dedicated; secure, costly	Bank/defence data centre
Community	Same need wali kai organizations share; cost shared	Govt. departments, hospitals
Hybrid	Public + private; sensitive data private mein	Payment private, website public

Pros and Cons of Cloud Computing

Pros: low cost (pay-per-use), scalability/elasticity, anywhere access, auto updates, easy backup & recovery.

Cons: internet dependency, security/privacy risk, limited control, vendor lock-in, downtime risk.